ALGEBRA

github.com/danimalabares/stack

Contents

1. Rings

Theorem 1.1. *Let R be a ring and $I \subset R$ an ideal. There is a correspondence between ideals of R containing I and ideals of R/I.*

Proof. We show that there are two maps between the set of ideals of R containing I and the ideals of R/I whose compositions are the identities in the corresponding set. One map is the projection to the quotient, and the other map is taking the elements whose equivalence class lies in a given ideal.

Suppose that $\tilde{J}$ is an ideal of R/I. Define $J = \{j \in R : j + I \in \tilde{J}\}$, the set of elements that project to $\tilde{J}$. Then $I \subset J$, because $i + I = I$ is the zero of R/I which is contained in any ideal of R/I. Also notice that J is an ideal of R since for any $r \in R$, $rj + I \in \tilde{J}$ by $\tilde{J}$ being an ideal. Notice that projecting J back to R/I gives J by definition.

Conversely, we project to the quotient. Let J be any ideal of R containing I Let $\tilde{J}$ be the projection to R/I. Then $\tilde{J}$ is an ideal of R/I by J being an ideal of R. Consider the set J' of all elements whose projection lies in $\tilde{J}$. Then $J' = J$ by definition. $\square$

This is the essential tool in proving that R/I is a field when I is a maximal ideal

Lemma 1.2. *Let R be a ring and I a maximal ideal of R. Then R/I is a field.*

Proof. It's easy to prove that a ring is a field if and only if its only ideals are $\{0\}$ and R. By Theorem 1.1, the ideals of R/I are in correspondence with ideals of R containing I, but only R is such. $\square$

2. Localization

The general recipe for localization is that $\mathcal{O}_U = \mathcal{O}_X[f^{-1}]$ where $U = X \setminus V(f)$. That is, localizing gives the sheaf of the variety without the zero set.

Definition 2.1. Let R be a ring and $S \subset R$ a multiplicative subset of R (containing 1). The *localization* of R at S is the ring RS^{-1} (where S^{-1} is just a set with the same elements as S but with a -1 exponent) with the operations of $(rs^{-1})(r_1 s_1^{-1}) = rr_1 s^{-1} s_1^{-1}$, $rs^{-1} + r_1 s_1^{-1} = (rs_1 + r_1 s)(s^{-1} s_1^{-1})$ and the identification given by $rs^{-1} = r_1 s_1^{-1} \iff rs_1 = r_1 s$. (Check this definition if R has zero divisors!)

In a simple case, consider a ring R and $f \in R$. The *localization of R at f* is $R[t]/(ft-1) := R[f^{-1}] = R(f)$. The quotient basically says that t is the inverse of f.

Example 2.2.
- Dyadic numbers: $\mathbb{Z}[2^{-1}]$, which are fractions of integers with denominators which are powers of 2.
- Laurent polynomials: $\mathbb{C}[t, t^{-1}]$.

Lemma 2.3. *If $R[f^{-1}] = 0$ then f is nilpotent.*

Proof. If $R[f^{-1}] = 0$ then $1 \in (ft-1)$, which means that $1 = (1-ft)p(t)$ for some $p(t) \in k[t]$. Then... □

The localization a ring is not always a local ring. But if S is a prime ideal, then it is.

Exercise 2.4. Prove that localization of a Noetherian ring is Noetherian.

Proof. Let R be a Noetherian ring and $S \subset R$. Let $I_1 \subset I_2 \subset \dots$ be a chain of ideals in the localization $R[S^{-1}]$. Let $\pi : R \to R[S^{-1}]$ the canonical projection $r \mapsto r$.

Let I be any of the I_i. Consider $\pi^{-1}(I)$. This gives a chain of ideals in R, which must stabilize. To go back to the localization simply recall that $\pi^{-1}(I)[S^{-1}] = I$. □

3. Noetherian and Artinian rings

Definition 3.1. A ring is called *Noetherian* if any increasing chain of ideals $I_1 \subset I_2 \subset \dots$ stabilizes. It is called *Artinian* if any descending chain of ideals $I_1 \supset I_2 \supset \dots$ stabilizes.

Exercise 3.2. A ring is Noetherian if and only if all its ideals are finitely generated.

Proof. First we prove that if not every ideal is finitely generated then R cannot be Noetherian. Indeed, if I is not finitely generated we construct an ascending chain by of ideals by starting with any generator of I and adding other generators one by one.

For the converse suppose that all ideals of R are finitely generated and pick an ascending chain $I_1 \subset I_2 \subset \dots$. Consider $I = \bigcup_i I_i$, which is an ideal since for any $a \in I$ there is i such that $a \in I_i$ so that $ra \in I_i$ for any $r \in R$. However, since it is finitely generated, all the generators have to be in some I_j, and thus the chain stabilizes. □

Theorem 3.3 (Hilbert basis theorem)**.** *If R is Noetherian then so is $R[x]$.*

Exercise 3.4.

Proof. $k[x_1, x_2, \dots]/(x_1^2, x_2^2, \dots)$, $I_n = (x_n, x_{n+1}, \dots)$ then $I_1 \supset I_2 \supset \dots$. □

Exercise 3.5. If A is Noetherian then so is $A[[t]]$.

Proof. Pass from an ideal in $A[[t]]$ to A by taking the coefficients of the monomials of minimal degree of each element in the ideal. Then take a finite set of generators. Then go back to $A[[t]]$ by taking the polynomials where the generators where taken. Then... □

4. Fields

Just for the record, any simple field extension, i.e. the smallest field in F containing a single element, is isomorphic to either que "rational function field $k(t)/k$" or to one of the field extensions $k[t]/(P)$ with $P \in k[t]$ irreducible. (Stacks Project tag 09G1.)

5. Modules

It looks like most books (at least [**?**] and [**?**]) define a module to be an abelian group along with a certain operation with elements of the ring.

Which secretly just says that

Definition 5.1. Let R be a ring. An *(left) R-module M* is an abelian group such that there exists a (left) ring morphism

$$R \to \mathrm{End}(M).$$

Indeed, the three usual requirements correspond to:

(1) The endomorphism corresponding to $a \in R$ respect the group structure of the group M:
$$a(x+y) = ax + ay,$$
(2) The representation map is a map of rings
$$(a+b)x = ax + by, \qquad (ab)x = a(bx).$$

6. Algebras

Definition 6.1. If R is a commutative ring, then a *commutative algebra* over R is a commutative ring S together with a ring morphism $R \to S$.

But actually I was expecting that S would be defined as a ring that is also an R-module. So let us note that a morphism of rings $R \to S$ gives a representation $R \to \mathrm{End}(S)$ via left multiplication. But the other way around, given an endomorphism associated to some element in R, how do we assign an element of S so as to produce a map $R \to S$?

7. Finitely-generated and finitely presented algebras

See Stacks Project tag 00F2.

Upshot: a finitely-generated R-algebra S is such that $S \simeq R[x_1, \dots, x_n]/I$ for some ideal $I \subset R$. Another way of saying this is that there is a surjective morphism $R[x_1, \dots, x_n] \to S$. Finitely-presentedness is when $I = (s_1, \dots, s_k)$.

As an aside, recall that finitely presented k-algebras with no nilpotents are in correspondence with affine algebraic varieties.

Definition 7.1. Let $R \to S$ be a ring map.

(1) We say $R \to S$ is of *finite type*, or that S *is a finite type R-algebra* if there exist an $n \in \mathbf{N}$ and a surjection of R-algebras $R[x_1, \dots, x_n] \to S$.
(2) We say $R \to S$ is of *finite presentation* if there exist integers $n, m \in \mathbf{N}$ and polynomials $f_1, \dots, f_m \in R[x_1, \dots, x_n]$ and an isomorphism of R-algebras $R[x_1, \dots, x_n]/(f_1, \dots, f_m) \cong S$.

For modules, the definition of finitely generated R-module is that there exists a surjective morphism $R^n \twoheadrightarrow S$. So: f.g. algebra means elements are polynomials on the generators with coefficients in the ring, and f.g. module means elements are linear combinations on the generators with coefficients in the ring.

8. Tensor product

Definition 8.1.

Theorem 8.2 (Universal property of tensor product)**.** *Let M_1, M_2, M be modules over a ring R. For any bilinear map $B : M_1 \times M_2 \to M$ there exists a unique homomorphism $b : M_1 \otimes M_2 \to M$ making the following diagram commute*

$$\begin{tikzcd} M_1 \times M_2 \arrow[r, "\pi"] \arrow[dr, "B"'] & M_1 \otimes M_2 \arrow[d, "b"] \\ & M \end{tikzcd}$$

where $\pi(m_1, m_2) = m_1 \otimes m_2$.

Remark 8.3. For $R = k$ the above property shows that $\mathrm{Bil}(M_1 \times M_2, k) = (M_1 \otimes M_2)^*$. If we assume that the vector spaces are finite dimensional, we can take duals to obtain

$$M_1 \otimes M_2 = (M_1 \otimes M_2)^{**} = \mathrm{Bil}(M_1 \times M_2, k)^*,$$

which is how Whitney first defined tensor products.

Here's Misha's way to show that the universal property of tensor product implies its uniqueness. Consider the category $\mathcal{C}$ of pairs of an R-module M, and a bilinear map $M_1 \times M_2 \to M$. Then we prove that the tensor product $M_1 \otimes M_2$ is an initial object in such category, and it is easy to show that initial objects are unique.

Remark 8.4. Misha's alternative way of stating the universal property of tensor product is that there is a bijective correspondence

$$\mathrm{Hom}_R(M_1 \otimes_R M_2, M) \leftrightsquigarrow \mathrm{Bil}(M_1 \times M_2, M).$$

The key observation is that the LHS are *linear* maps from a module to M while the RHS are *bilinear* maps.

Exercise 8.5 (Currying)**.**

$$\mathrm{Hom}(M_1 \otimes_R M_2, M) = \mathrm{Hom}(M_1, \mathrm{Hom}(M_2, M))$$

Proof. This is implied by the universal property since the RHS is just $\mathrm{Bil}(M_1 \times M_2, M)$. ☐

Exercise 8.6. Let

$$M_1 \xrightarrow{f_1} M_2 \xrightarrow{f_2} M_3 \longrightarrow 0$$

be exact. Show that

$$0 \longrightarrow \mathrm{Hom}(M_3, N) \xrightarrow{-\circ f_2} \mathrm{Hom}(M_2, N) \xrightarrow{-\circ f_1} \mathrm{Hom}(M_1, N)$$

is exact.

Proof. Injectivity of the first arrow is easy. For the middle term pick $f \in \mathrm{Hom}_R(M_2, N)$ in the kernel of $- \circ f_1$. To show that it is in the image of $- \circ f_2$ we look for $g \in \mathrm{Hom}_R(M_3, N)$ such that $f = g \circ f_2$.

Since $M_1 \to M_2 \to M_3 \to 0$ is exact, we have an isomorphism $\bar{f}_2 : M_2/\mathrm{img} f_1 \xrightarrow{\simeq} M_3$. Since $f \circ f_1 = 0$, we see that f induces a map $\bar{f}$ on the quotient since Indeed, if $m_2 - m_2' \in \mathrm{img} f_1$, we have $0 = f(m_2 - m_2') = f(m_2) - f(m_2')$.

Then for any $m_3 \in M_3$ we have a unique class $m_2 + M_1 \in M_2/\mathrm{img} f_1$. Thus we can define $g = \bar{f} \circ \bar{f}_2^{-1}$.

Note that this choice satisfies $f = g \circ f_2$. Indeed,

$$g \circ f_2(m_2) = \bar{f} \circ \bar{f}_2^{-1}(m_3) = \bar{f} \circ (m_2 + M) = f(m_2).$$

☐

Exercise 8.7. Let $R = \mathbb{Z}$. Find an example of an injective homomorphism $M_1 \to M_2$ such that $\mathrm{Hom}(M_2, N) \to \mathrm{Hom}(M_1, N)$ is not surjective.

Proof. Consider

$$\begin{tikzcd}
0 \arrow[r] & 2\mathbb{Z} \arrow[r, "i"] \arrow[d, "\varphi"'] & \mathbb{Z} \arrow[dl, dashed, "\nexists"] \\
 & \mathbb{Z}/2\mathbb{Z} &
\end{tikzcd}$$

where

$$\begin{aligned} \varphi : 2\mathbb{Z} &\longrightarrow \mathbb{Z}/2\mathbb{Z} \\ 2n &\longmapsto [n]. \end{aligned}$$

Then there is no extension $\bar{\varphi} : \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ making $\varphi = \bar{\varphi} \circ i$. Indeed, note that $\varphi(2) = [1]$, while $\bar{\varphi}(2 \cdot 1) = 2\bar{\varphi}(1) = [0]$. ☐

The idea is that any element in the image must be divisible by any integer. This is why $\mathbb{Q}$ is an injective module according to the following definition. (A precise proof of why $\mathbb{Q}$ is injective can be found at 01D6.)

Definition 8.8. An R-module is *injective* if the functor $\operatorname{Hom}(-, M)$ is exact. That is, if the dotted arrow in the following diagram always exists:

$$\begin{tikzcd}
0 \arrow[r] & M_1 \arrow[r, hook] \arrow[d] & M_2 \arrow[dl, dashed, "\exists"] \\
& N &
\end{tikzcd}$$

Exercise 8.9. Let

$$0 \longrightarrow M_1 \xrightarrow{f_1} M_2 \xrightarrow{f_2} M_3 \longrightarrow 0$$

be an exact sequence of R-modules. Prove that

$$0 \longrightarrow \operatorname{Hom}_R(N, M_1) \xrightarrow{f_1 \circ -} \operatorname{Hom}_R(N, M_2) \xrightarrow{f_2 \circ -} \operatorname{Hom}_R(N, M_3)$$

is exact

Proof. Injectivity is simple. For the middle term let $f : N \to M_2$ be in the kernel of $f_2 \circ -$, that is, $f(N)$ is killed by f_2.

We want to define a map $g : N \to M_1$ such that $f_1 \circ g = f$. But observe that the image of f already lies in $M_1 = \operatorname{Ker} f_2$. Thus f is already a map $f : N \to M_1$, save a more precise notation. □

Exercise 8.10. Let $R = \mathbb{Z}$. Find an epimorphism (surjective homomorphism) of R-modules $M_2 \to M_3$ such that the corresponding map

$$\operatorname{Hom}_R(N, M_2) \to \operatorname{Hom}_R(N, M_3)$$

is not surjective for some R-module N.

Proof. Now consider

$$\begin{tikzcd}
& \mathbb{Z} \arrow[dl, dashed, "\nexists"'] \arrow[d, "\pi"] & \\
2\mathbb{Z} \arrow[r, "\varphi"'] & \mathbb{Z}/2\mathbb{Z} \arrow[r] & 0
\end{tikzcd}$$

for the same φ as in Exercise 8.7. Then $\pi(1) = [1]$, while any extension mapping $1 \to 2n$ gives $\varphi(2n) = 2\varphi(n) = [0]$. □

Definition 8.11. An R-module N is *projective* if the functor $\operatorname{Hom}_R(N, -)$ is exact. That is, if the dotted arrow in the following diagram always exists:

$$\begin{tikzcd}
& N \arrow[dl, dashed, "\exists"'] \arrow[d] & \\
M_2 \arrow[r, two heads] & M_3 \arrow[r] & 0
\end{tikzcd}$$

Exercise 8.12. A finitely generated R-module N is projective if and only if it is a direct summand of a free R-module R^n, i.e. there is a direct sum decomposition $R^n = N \oplus N'$.

Proof. Consider $R^n \simeq \bigoplus_{i=1}^n a_i R$ where N is generated by $a_1, \dots, a_n$. Apply projectivity to obtain

$$\begin{tikzcd} & N \arrow[dl, dashed, "\exists\psi"'] \arrow[d, "\mathrm{id}"] & \\ R^n \arrow[r, "\varphi"'] & N \arrow[r] & 0 \end{tikzcd}$$

where

$$\varphi : R^n \longrightarrow N$$
$$(r_1 a_1, \dots, r_n a_n) \longmapsto \sum r_i a_i.$$

This is a section of φ in the exact sequence

$$\begin{tikzcd} 0 \arrow[r] & \operatorname{Ker}\varphi \arrow[r] & R^n \arrow[r, "\varphi"] & N \arrow[l, bend left, "\psi"] \arrow[r] & 0 \end{tikzcd}$$

which gives decomposition

$$R^n \simeq \operatorname{Ker}\varphi \oplus \mathrm{Im}\psi.$$

More explicitly, the intersection of those modules is clearly trivial while any element of R^n may be written as

$$x = \underbrace{(x - \psi\varphi(x))}_{\in \operatorname{Ker}\varphi} + \underbrace{\psi\varphi(x)}_{\in \mathrm{Im}\psi}.$$

For the converse we start from the diagram given in the definition of projective module, add N' to every term, arriving at $R^n \simeq N \oplus N'$. We observe that free modules are projective by constructing an extension using a basis of R^n, and then back to N by taking the action of the extension restricted to N. $\square$

In fact, currying (Exercise 8.5) is the essential ingredient for the proof that tensor product is left exact. Such result follows from two lemmas.

Lemma 8.13. *Let*

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

be an exact sequence of R-modules. Then for any R-modules N and N', the sequence

$$0 \longrightarrow \mathrm{Hom}_R(M_3 \otimes N', N) \longrightarrow \mathrm{Hom}_R(M_2 \otimes N', N) \longrightarrow \mathrm{Hom}_R(M_1 \otimes N', N)$$

is exact.

Proof. Apply $\mathrm{Hom}_R(-, N)$, then $\mathrm{Hom}_R(N', -)$ and then currying. $\square$

There is an (almost) converse to Exercise 8.6:

Lemma 8.14. *Let*

$$E : \qquad M_1 \xrightarrow{\mu} M_2 \xrightarrow{\rho} M_3 \longrightarrow 0$$

be a complex, i.e. $\rho\mu = 0$, such that

$$0 \longrightarrow \mathrm{Hom}_R(M_3, N) \xrightarrow{\rho_N} \mathrm{Hom}_R(M_2, N) \xrightarrow{\mu_N} \mathrm{Hom}_R(M_1, N)$$

is exact for any R-module N. Then E is also exact.

And this is what implies right-exactness of tensor product:

Theorem 8.15. *Let*

$$M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

be exact. Then

$$M_1 \otimes_R M \longrightarrow M_2 \otimes_R M \longrightarrow M_3 \otimes_R M \longrightarrow 0$$

is exact.

Proof. This is just Lemma 8.13 and Lemma 8.14!!! □

Exercise 8.16. Let $R = \mathbb{C}[t]$. Find an exact sequence of R-modules

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

and an R-module N such that the corresponding map

$$M_1 \otimes_R N \to M_2 \otimes_R N$$

is not injective.

Proof. Suppose the given exact sequence is just the ideal exact sequence of a point in $\mathbb{A}^1$. Then tensoring by M means thickening the point. Not having injectivity after tensoring means that tensoring the ideal by M does not give the ideal of the resulting variety. □

And from this it follows as corollary

Proposition 8.17. *Let $I \subset R$ be an ideal of a ring. Then $M \otimes_R (R/I) = M/IM$.*

Proof. Take the exact sequence $0 \to I \to R \to R/I$ and tensor by M. □

Exercise 8.18. Let I, I' be distinct maximal ideals in a ring R. Then $R/I \otimes_R R/I' = 0$.

Proof. They key observation is that since I and I' are maximal we must have $I + I' = M$. This means that $1 = f + f'$ for some $f \in I$ and $f' \in I'$. Then $1(a+I) \otimes (b+I') = (fa+I) \otimes (b+I') + (a+I) \otimes (fb+I') = 0$. □

Now we discuss tensor product in the category of rings over a ring.

Definition 8.19. Let A, B and C be rings, and $C \to A$ and $C \to B$ homomorphisms. Considering A and B as C-modules, we can define a product on $A \otimes_C B$ by

$$a \otimes b \cdot a' \otimes b' := aa' \otimes bb'.$$

By the universal property this indeed extends to a product.

Example 8.20.

$$\mathbb{C}[t_1, \ldots, t_k] \otimes_{\mathbb{C}} \mathbb{C}[z_1, \ldots, z_n].$$

Example 8.21. For any homomorphism $\varphi : C \to A$, the ring $A \otimes_C (C/I)$ is isomorphic to $\frac{A}{\varphi(I)A}$ by Proposition 8.17.

Recall that the spectrum of a ring $\operatorname{Spec} R$ is the corresponding algebraic variety.

Proposition 8.22. *Let $f : X \to Y$ be a morphism of affine varieties and $f^* : \mathcal{O}_Y \to \mathcal{O}_X$ the corresponding morphism. Let $y \in Y$ and $\mathfrak{m}_y$ the corresponding maximal ideal. Then*

$$\operatorname{Spec}((\mathcal{O}_X \otimes_{\mathcal{O}_Y} \frac{\mathcal{O}_Y}{\mathfrak{m}_y}) \Big/ \textit{nilradical}) \cong f_1(y).$$

Lemma 8.23. *Let A, B be finitely generated reduced $\mathbb{C}$-algebras. If $R := A \otimes_{\mathbb{C}} B$, and $N \subset R$ nilradical (the set of nilpotent elements of R). Then*

$$\operatorname{Spec}(R/N) = \operatorname{Spec}(A) \times \operatorname{Spec}(B).$$

Now we discuss...

Lemma 8.24. *Let A be a finitely-generated $\mathbb{C}$-algebra. The intersection of all the prime ideals of A is its nilradical.*

Exercise 8.25. Let A and B be finitely generated $\mathbb{C}$-algebras, $B \to A$ a homomorphism and $\mathfrak{m} \subset B$ a maximal ideal. Then $A \otimes_B (B/\mathfrak{m})$ can contain nilpotents.

Proof. The geometric idea is to project a parabola to an axis. At the point of tangency we obtain a fat point, whose algebra is not reduced. □

However, over $\mathbb{C}$ the situation is different:

Theorem 8.26. *Let A, B be finitely generated reduced $\mathbb{C}$-algebras. Then $A \otimes_{\mathbb{C}} B$ is reduced.*

It follows that

Theorem 8.27.

$$\operatorname{Spec}(A) \times \operatorname{Spec}(B) = \operatorname{Spec}(A \otimes_{\mathbb{C}} B).$$

Now we can take preimages!

Lemma 8.28. *Let $f : X \to Y$ be a morphism of affine algebraic varieties, $f^* : \mathcal{O}_Y \to \mathcal{O}_X$ the corresponding ring homomorphism, $Z \subset Y$ a subvariety and I_Z its ideal. Denote by R_1 the quotient ring*

$$(\mathcal{O}_X \otimes_{\mathcal{O}_Y} (\mathcal{O}_Y/IZ)/nilradical = (\mathcal{O}_X/f^*(I_Z))/nilradical.$$

Then

$$\operatorname{Spec}(R_1) = f^{-1}(Z).$$

Proof. Analogous to preimage of a point. □

The slogan is: adding polynomials to an ideal is the same as taking tensor product. This operation is given by Exercise 8.18.

We can also consider the diagonal of a product:

Lemma 8.29. *Let M be an affine algebraic variety, its diagonal $\Delta \subset M \times M$ and $I \subset \mathcal{O}_M \otimes_{\mathbb{C}} \mathcal{O}_M$ the ideal generated by $r \otimes 1 - 1 \otimes r$ for all $r \in \mathcal{O}_M$. Then*

$$\mathcal{O}_M \otimes_{\mathbb{C}} \mathcal{O}_M/I.$$

And now we introduce fibered product!

Proposition 8.30. *Let $X \to M$ and $Y \to M$ be morphisms of algebraic varieties, $R = \mathcal{O}_X \otimes_{\mathcal{O}_M} \mathcal{O}_Y$, and R_1 the quotient of R by its nilradical. Then*

$$\operatorname{Spec}(R_1) = X \times_M Y.$$

Proof.

$$X \times Y \xrightarrow{f \times g} Z \times Z,$$

and compute $(f \times g)^{-1}(\Delta)$. By Lemma [?] we must have

$$A \otimes_{\mathbb{C}} B / (r \otimes 1 - 1 \otimes r) = A \otimes_C B$$

where $A = \mathcal{O}_X$, $B = \mathcal{O}_Y$ and $C = \mathcal{O}_Z$. □

We shall not explain why, but coproduct in the category of reduced finitely generated algebras is direct sum.

9. Artinian algebras

We start with a motivating introduction. Field of rational functions on a variety, take its covering, remove its ramification locus, obtain a field extension. Study automorphisms of the field extensions, which is the same as studying automorphisms of the cover. This is Galois theory. Decomposition of a direct sum of fields shall correspond to the disjoint union of varieties.

In this section algebras are not assumed to have unity.

A *field extension* of the field k is another field K which cointains k. This is denoted $[K : k]$.

$[K : k]$ is *algebraic* if K is finite-dimensional over k. Otherwise it is *trascendental.*

An element $x \in K$ is *algebraic* if it is a root of a nonzero polynomial with coefficients in k, and otherwise it is *trascendental.*

Exercise 9.1. A field extension is algebraic if and only if all its elements are algebraic.

A field is called *algebraically closed* is all algebraic extensions are trivial. This is equivalent to the field containing the roots of all polynomials.

Definition 9.2. The *minimal polynomial* of an element v in a finitely-generated k-algebra R a monic polynomial $P \in R[t]$ of smallest possible degree having v as a root.

Lemma 9.3. *The subalgebra generated by v is isomorphic to $k[t]/(P)$.*

Proof. Consider the map $k[t] \to Rv$, whose kernel is (P) since P divides any polynomial vanishing in v. □

For example,

$$\frac{\mathbb{R}[t]}{t^2+1} \simeq \mathbb{C}$$

while

$$\frac{\mathbb{R}[t]}{t^2=1} \simeq \mathbb{R} \oplus \mathbb{R}.$$

Recall that $k[t]$ is a UFD. Also recall that P is an irreducible element of $k[t]$ if and only if $k[t]/(P)$ is a field. This follows since $k[t]/(P)$ cannot have zerodivisors if P is prime, which is equivalent to irreducible in UFDs. These extensions are called *primitive*. [They are extensions of k obtained by adding a root of P.]

Definition 9.4. A commutative, associative k-algebra R not necessarily with unit is called an *Artinian algebra* if it is a finite-dimensional vector space over k. It is called *semisimple* if it has no nonzero nilpotents.

We will show that any semisimple Artinian algebra is a direct sum (direct sum of rings equipped with natural operations) of fields, and this decomposition is uniquely defined.

An idempotent element e in R, i.e. $e^2 = e$, gives us a canonical decomposition

$$eR \oplus (1-e)R$$

since $1 - e$ is also idempotent.

Theorem 9.5. *An Artinian algebra without nilpotents contains an idempotent.*

The proof of the theorem constructs a maximal ideal $I \subset A$ which contains a unit, and thus is a field I.

Thus we obtain

Theorem 9.6. *An Artinian algebra with unity with no nilpotents is a direct sum of fields.*

Proof. We find an idempotent element e, so that we can decompose

$$A = eA \oplus (1-e)A$$

where e is a unity on the first factor and so is $1-e$ on the second factor, thus every component is a field. We repeat this process until we exhaust the dimension of A. □

We can also prove this theorem for algebras without unity. We consider the kernel of the map $u_0 : a \mapsto au$ and put

$$A = uA \oplus \operatorname{Ker} u_0 A.$$

Lemma 9.7. *Any decomposition of an algebra into a direct sum of fields is unique up to permutation of the factors.*

10. Finite morphisms

Remark 10.1. If M is a finitely generated R-module and $R \to R'$ is a ring morphism, then $M \otimes_R R'$ is a finitely-generated R'-module generated by the same generators of M.

Definition 10.2. A *finite morphism* of affine varieties is called *finite* if the $\mathcal{O}_X$ is a finitely generated $\mathcal{O}_Y$-module. In this case, $\mathcal{O}_X$ is called an *integral extension* of $\mathcal{O}_Y$.

Here's some notes from the past that may pop up in memory: The upshot is: let $\varphi : R \to S$ be a ring map. An element $s \in S$ is integral over R if there is a monic polynomial with coefficients in R that gives zero when you put s instead of the variable.

But it might be convenient to think about finite (or, for humans, finitely-generated) R-modules: x is integral over R if and only if $R[x]$ is a finitely-generated R-module (i.e., not only finitely generated as an algebra but also as a module). See Zvi Rosen's video. The forward implication is: suppose x is integral, then $x^n + r_1x^{n-1} + \ldots + r_n = 0$, so $x^n = -r_1x^{n-1} + \ldots + r_n$, which "says" $R[x] \cong \bigoplus_{k=0}^{n-1} Rx^k$.

So maybe the details of that are not entirely trivial (Zvi uses some theorem, and Stacks Project does lots of things). But the point is: I think that after applying Spec, the fibers of an integral extension are **finite**. Which of course has all to do with the finite-moduleness I was trying to explain.

Back to Misha's lecture…

Theorem 10.3.

Proof. The decomposition into sums of fields gives the finiteness. □

Now we discuss the trace form.

Definition 10.4. Let R be a k-algebra, and $g : R \times R \to k$ a symmetric k-bilinear form on R. g is called *invariant* if

$$g(x, yz) = g(xy, z).$$

If R has a unity we have

$$g(x, y) = g(xy, 1),$$

so that in fact g is determined by a linear functional $a \mapsto g(a, 1)$.

Example 10.5. In the ring $\mathbb{C}[x, y]/(x^{n+1}, y^{n+1})$ the form $\varepsilon(xy) = a_n n$ is nondegenerate.

Definition 10.6 (Trace form. . .)**.**

Definition 10.7. A field extension is *separable* if the trace form is nonzero.

Theorem 10.8. *Let R be an Artinian algebra over k with nondegenerate trace form. Then R is semisimple.*

11. Gröbner theory

So far here's some rather unformatted notes from Lin's talk at Commutative Algebra school in Recife 2025.

Would SAGBI bases be related to this?

square free monomial ideals preserve regularity upon Grobner deformation i.e. initial ideal!

Constantini-Fouli-Goel-Lin-Lindo-Liske-Mostafazadehfard, 2025.

https://sites.psu.edu/kul20/

How is Lin using the Reese algebra here? Is it ture that the initial ideal of the Gröbner deformation coincides with the initial ideal "of the Resse algebra"?

References